# Computer Crime - Self Study Note for Students

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Based on Week 5, Lecture 1 & Lecture 2
Reference Book : A Gift of Fire, Chapter 2

The definitions and topics follow lecture slides, with additional scenarios to make the important concepts easier to understand.

## 1. Computer Crime

Definition: Computer crime refers to illegal activities in which a computer is used as a tool to commit a crime or becomes the target of a crime.

Example: A criminal gains unauthorized access to a university database and steals students' personal information.

## 2. Three Major Challenges of Computer Crime

The three main challenges are Prevention, Detection, and Prosecution.

***1. Prevention***

Definition: Prevention means taking security measures to stop computer crimes before they happen.

Example: Using strong passwords, antivirus software, firewalls, and regular security updates.

Scenario: A university uses strong passwords and two-factor authentication to protect student accounts. A hacker tries to log in using a stolen password but cannot access the account because additional verification is required.

***2. Detection***

Definition: Detection means identifying computer crimes, suspicious activities, or security breaches after they occur or while they are happening.

Example: Using security monitoring software, intrusion detection systems, and activity logs.

Scenario: A bank's security system detects several unusual login attempts from an unknown location. The system alerts the security team, which investigates and discovers an attempted cyberattack.

***3. Prosecution***

Definition: Prosecution is the legal process of investigating, charging, and bringing cybercriminals to court for their illegal activities.

Example: Police use digital evidence, such as server logs and transaction records, to investigate a cybercriminal.

Scenario: A hacker steals money from an online banking account. Investigators collect digital evidence and identify the suspect. The suspect is arrested and brought before a court for legal action.

Shortcut - Easy way to remember:

- Prevention = Stop the crime.
- Detection = Find the crime.
- Prosecution = Take legal action against the criminal.

## 3. Hacking and Its Three Phases

Definition of Hacking: Hacking refers to accessing or manipulating computer systems using technical knowledge. In the context of computer crime, it usually means gaining unauthorized access to computers, networks, or data.

### Phase 1: Joy of Programming (1960s–1970s)

- Hacking was originally considered a positive activity.
- Hackers were creative programmers who developed clever programs.
- They helped create early operating systems and computer games.
- Many hackers were high-school or college students.

### Phase 2: Negative Meaning (1970s–Mid-1990s)

- Hacking began to be associated with unauthorized computer access.
- Criminals targeted business and government computer systems.
- Computer-related white-collar crimes increased.
- Organized criminal groups became involved in hacking.

### Phase 3: Growth of the Web and Mobile Devices (Mid-1990s onward)

- Internet growth created more opportunities for cybercriminals.
- Attackers began spreading malicious software.
- Unprotected computers became easier targets.
- Politically motivated hacking increased.
- Script kiddies could carry out attacks using tools created by others.

## 4. Hacktivism (***)

**Definition:** Hacktivism is the use of hacking techniques to promote a political or social cause.

Common activities:

- Website defacement: Changing a website's content without permission.
- Data destruction: Deleting or damaging information.
- Denial-of-Service (DoS): Making a website or service unavailable to its users.

Scenario: A group of activists disagrees with a government policy. They hack a government website and replace its homepage with a protest message to express their political opinion.

Important point: Hacktivism is controversial. Some consider it digital activism or civil disobedience, while others consider it cybercrime because it can violate property rights and disrupt services.

## 5. Cybercrime

Definition: Cybercrime is any criminal activity involving a computer, networked device, or computer network. It may involve stealing information, damaging systems, committing fraud, or harming individuals through the internet.

Note: Cybercrime is a subset of computer crime.

### Targets of Cybercrime

| Target | Common crimes |
|---|---|
| Individuals and their property | Virus attacks, unauthorized access, intellectual property theft |
| Organizations | Confidential information theft, cyberterrorism, software piracy |
| Society at large | Financial fraud, illegal online gambling, trafficking, forgery, sale of illegal goods |

## 6. Common Types of Cybercrime

### 1. Hacking (***)

**Important**

Definition: Hacking is gaining unauthorized access to a computer system or network to steal, modify, or misuse information.

Example: Breaking into a university's database without permission.

Scenario: A student uses stolen administrator credentials to enter the university's database. The student changes examination marks without permission. This is hacking because the student accessed and modified information illegally.

### 2. Phishing (***)

**Important**

Definition: Phishing is a cybercrime in which criminals send fake emails or messages containing fraudulent links to trick people into revealing sensitive information, such as passwords or bank details.

Example: A fake bank email asking a customer to verify their account.

Scenario: A person receives an email that appears to come from their bank. It says their account will be blocked unless they click a link and log in. The person enters their password on the fake website, allowing the criminal to steal the login information.

### 3. Spoofing (***)

**Important**

Definition: Spoofing is a technique in which an attacker pretends to be a trusted person, organization, website, or device to deceive users, steal data, or gain unauthorized access.

Example: A criminal sends an email that appears to come from a company's official email address.

Scenario: An employee receives an email that appears to be from their company CEO, requesting an urgent money transfer. The attacker has forged the sender's identity. Believing the email is genuine, the employee transfers money to the criminal.

### 4. Computer Virus

Definition: A computer virus is a malicious program that attaches itself to other programs or files, spreads when they are executed, and can damage data or computer systems.

### 5. Netspionage (Network Espionage)

Definition: Netspionage is the unauthorized access to computer systems or networks to obtain confidential information, often for the purpose of selling it or sharing it with others.

### 6. Cyberstalking

Definition: Cyberstalking is the repeated use of the internet, email, social media, or other electronic communication systems to harass, threaten, or monitor another person.

### 7. Cyberterrorism

Definition: Cyberterrorism refers to unlawful attacks or threats against computer systems and networks intended to intimidate or coerce a government or population for political or social purposes.

## 7. Common Types of Cybercriminals

| Type | Definition |
|---|---|
| Black Hat Hackers | Individuals who illegally access computer systems to steal information, cause damage, or gain financial benefits. |
| Script Kiddies | Inexperienced attackers who use existing hacking tools developed by others. |
| Organized Cybercriminals | Groups that carry out coordinated cybercrimes, usually for financial profit. |
| Cyberterrorists | Individuals or groups who use cyberattacks to intimidate governments or societies for political or ideological purposes. |
| Hacktivists | Individuals who use hacking techniques to support political or social causes. |
| Malicious Insiders | Employees or other trusted individuals who misuse their authorized access to steal data or damage systems. |

## 8. Motives of Cybercriminals

1. Entertainment: Attacking computer systems for amusement or excitement.
2. Financial Profit: Stealing money or selling confidential information.
3. Anger or Revenge: Attacking someone because of personal conflict.
4. Political Agenda: Using cyberattacks to support political goals.
5. Sexual Motivation: Committing online crimes involving sexual exploitation or harassment.
6. Psychological or Psychiatric Factors: Psychological motivations or, in some cases, mental illness may play a role; mental illness does not itself imply criminal behavior.

## 9. Prevention of Cybercrime

- Strong passwords: Use passwords that are difficult to guess.
- Antivirus software: Install and regularly update antivirus protection.
- Firewalls: Use firewalls to block unauthorized network access.
- Email safety: Avoid opening suspicious links or attachments from unknown sources.
- Data backup: Regularly back up important files to prevent permanent data loss.

Additional good practice includes using two-factor authentication, updating operating systems, and avoiding suspicious websites.

## 10. Important Differences

| Hacking | Phishing | Spoofing |
| --- | --- | --- |
| Unauthorized access to a system or data | Tricking users into revealing sensitive information | Pretending to be a trusted source or identity |
| Main focus: Access | Main focus: Deception to steal information | Main focus: Impersonation |
| Example: Breaking into a database | Example: Fake bank login email | Example: Forged CEO email address |

Note: Phishing and spoofing can occur together. For instance, a phishing email may use email spoofing to appear as though it came from a legitimate bank.